Product PS: MediBot

Design and build MediBot, an intelligent conversational assistant that helps users manage their medication needs while strictly adhering to medical safety and regulatory guidelines.

Submit Solution | Evaluation Rubric

01 Introduction & Problem Statement

The rapid growth of online pharmacies, coupled with rising healthcare costs, has made access to reliable medication information a pressing concern for consumers. Users find it difficult to compare prices across platforms, to identify safe and cost-effective alternatives, and to obtain dependable answers to general medication-related queries. There is currently no accessible, conversational, and safety-conscious interface that addresses these needs in one place.

Problem Statement:
Participants are required to design and build **MediBot**, an intelligent conversational assistant that helps users manage their medication needs while strictly adhering to medical safety and regulatory guidelines.

1.1 Core Challenges Addressed

1. Fragmented Information — E-Pharmacy Friction

Medicine prices and availability are dispersed across numerous e-pharmacy platforms, making the search for the best offer time-consuming.

2. Lack of Alternative Discovery — Cost Disparity

Users are frequently unaware of generic or equivalent medications with identical active ingredients available at a significantly lower cost.

3. Limited Accessibility of Information — 24/7 Availability

There is no reliable, round-the-clock conversational agent for basic medication queries, side-effect questions, or dosage instructions.

4. Safety and Compliance Risks — Critical Hazard

In the absence of safeguards, users may self-medicate with high-risk or prescription-only drugs, leading to potential health hazards.

1.2 System Objective

To develop a centralised, intelligent medical chatbot that:

- Provides accurate medicine price comparisons across certified e-pharmacies.
- Recommends composition-equivalent alternatives with identical active chemical ingredients.
- Answers general medication and health-information queries in natural language.
- Enforces strict prescription-safety controls at all times.

02 Target Users

MediBot addresses the distinct needs of three key consumer archetypes:

User 1

Chronic Patients

Individuals requiring regular, recurring medication refills who need continuous stock tracking, predictable pricing, and reliable delivery windows.

User 2

Budget-Conscious Consumers

Users seeking the most affordable options, verified generic alternatives, and transparent cross-platform price matching.

User 3

Information Seekers

Individuals looking for clear, reliable basic information on drug interactions, side effects, and safe usage guidelines.

03 Core Features & Architecture

The solution must implement four foundational functional modules:

FEATURE	DESCRIPTION
`Price Comparison`	Aggregates data from certified e-pharmacy sources and enables comparison of price, delivery time, and availability.
`Alternative Drugs`	Suggests cheaper generic or equivalent drugs with the exact same active composition.
`Health Assistant`	A natural language interface for general health queries, medication side effects, and usage instructions.

FEATURE	DESCRIPTION
`Prescription Guard`	Immediately identifies restricted / Schedule H drugs and enforces the mandated prompt.

Mandated Prescription Guard Intercept Prompt:
When a restricted or Schedule H drug is queried, the bot must immediately output:
`"This medication requires a valid prescription. Please consult your doctor."`

Interactive MediBot Query Sandbox

Simulate bot queries, Schedule H interception, composition matching, and price comparisons.

Select Query Preset: **Augmentin 625 Duo**

Augmentin 625 Duo (Schedule H) | Crocin 650 (OTC Analgesic) | Alprazolam 0.5mg (Restricted) | Allegra 120mg (Antihistamine)

⚠️ **Schedule H Intercept:** This medication requires a valid prescription. Please consult your doctor.

IDENTIFIED COMPOUND
Augmentin 625 Duo
Antibiotic / Penicillin Combination

ACTIVE COMPOSITION
Amoxicillin (500mg) + Clavulanic Acid (125mg)
Base MRP: ₹223.50

MAX POTENTIAL SAVINGS
32.3% Savings
2 Verified Alternative(s)

SUGGESTED EQUIVALENT	ACTIVE MATCH	MRP & SAVINGS	PARTNER PHARMACY QUOTES
Moxikind-CV 625 Mankind Pharma Ltd	100% Identical Active	₹151.20 **32.3%** **Savings**	1mg: ₹136.08 (Tomorrow by 2 PM) PharmEasy: ₹139.50 (In 24-48 hrs) Apollo 24|7: ₹142.00 (Same Day (2 hrs))

SUGGESTED EQUIVALENT	ACTIVE MATCH	MRP & SAVINGS	PARTNER PHARMACY QUOTES
Clavam 625 Alkem Laboratories Ltd	100% Identical Active	₹168.00 **24.8%** **Savings**	1mg: ₹149.00 (Tomorrow) PharmEasy: ₹152.00 (In 2 days) Netmeds: ₹150.50 (Tomorrow)

3.1 API Query & Response Schema

JSON **MediBot Query Response Schema (JSON)**

```
// Example MediBot Query & Substitution API: POST /api/v1/medibot/query
{
  "user_query": "Can I get a cheaper alternative to Augmentin 625 Duo?",
  "session_id": "sess_98412_a3",
  "disclaimer_acknowledged": true,
  "safety_guard": {
    "is_restricted_schedule_h": true,
    "intercept_triggered": true,
    "mandated_prompt": "This medication requires a valid prescription. Please consul
    "is_diagnosis_detected": false
  },
  "detected_medication": {
    "brand_name": "Augmentin 625 Duo",
    "active_composition": "Amoxicillin (500mg) + Clavulanic Acid (125mg)",
    "therapeutic_class": "Antibacterial / Penicillin Combination",
    "schedule": "Schedule H",
    "mrp_inr": 223.50
  },
  "substitutes": [
    {
      "brand_name": "Moxikind-CV 625",
      "manufacturer": "Mankind Pharma Ltd",
      "active_composition": "Amoxicillin (500mg) + Clavulanic Acid (125mg)",
      "composition_identically_verified": true,
      "mrp_inr": 151.20,
      "savings_percentage": 32.3,
      "user_advisory": "Active composition is verifiably identical. Please confirm a
      "partner_price_comparison": [
        { "platform": "1mg", "price_inr": 136.08, "delivery_eta": "Tomorrow by 2 PM"
        { "platform": "PharmEasy", "price_inr": 139.50, "delivery_eta": "24-48 hrs",
```

```
        { "platform": "Apollo 24|7", "price_inr": 142.00, "delivery_eta": "2 Hours (
      ]
    }
  ],
  "disclaimer": "MediBot is an informational tool. It does not diagnose medical cond
}
```

3.2 Guardrail & Substitution Module

PYTHON Prescription Guard & Substitution Verification Starter (Python)

```python
import re
from typing import Dict, Any, List

SCHEDULE_H_PATTERNS = [
    r"schedule[- ]?h", r"prescription[- ]?only", r"rx", r"antibiotic",
    r"alprazolam", r"amoxicillin", r"azithromycin", r"diazepam", r"clonazepam"
]

MANDATED_PRESCRIPTION_PROMPT = (
    "This medication requires a valid prescription. Please consult your doctor."
)
MANDATORY_DISCLAIMER = (
    "Disclaimer: MediBot is an informational tool. It does not diagnose medical cond
    "or prescribe medications. Users must always consult a qualified medical profess
    "before taking any medication."
)

def evaluate_prescription_guard(medication_record: Dict[str, Any]) -> Dict[str, Any]
    """
    Mandatory Prescription Guard:
    Intercepts 100% of Schedule H / restricted requests.
    Prevents unauthorized diagnosis and prescription.
    """
    raw_text = f"{medication_record.get('name', '')} {medication_record.get('schedul
    is_schedule_h = any(re.search(pat, raw_text) for pat in SCHEDULE_H_PATTERNS)

    return {
        "is_restricted": is_schedule_h,
        "mandated_prompt": MANDATED_PRESCRIPTION_PROMPT if is_schedule_h else None,
        "disclaimer": MANDATORY_DISCLAIMER
    }
```

```
def find_verified_substitutes(target_med: Dict[str, Any], catalog: List[Dict[str, An
    """
    Strict composition matching:
    Only suggest medicines where active composition is verifiably identical.
    """
    target_comp = target_med['active_composition'].strip().lower()
    verified_alts = []

    for item in catalog:
        if item['brand_name'] == target_med['brand_name']:
            continue
        if item['active_composition'].strip().lower() == target_comp:
            savings = round(((target_med['mrp'] - item['mrp']) / target_med['mrp'])
            verified_alts.append({
                "brand_name": item['brand_name'],
                "manufacturer": item['manufacturer'],
                "active_composition": item['active_composition'],
                "savings_pct": savings,
                "verified_identical": True,
                "advisory": "Active composition is verifiably identical. Confirm wit
            })
    return sorted(verified_alts, key=lambda x: x['savings_pct'], reverse=True)
```

04 Datasets & Resources

Participants may use the following datasets to map brand names to active ingredients and to build the price-comparison and substitution features.

4.1 Provided Datasets

DATASET	SOURCE	DESCRIPTION	ACCESS LINK
1mg Medicine Dataset (2024/2025)	Kaggle	Scraped data from 1mg containing drug name, manufacturer, MRP, chemical composition, and substitute options.	Kaggle 1mg Dataset ↗

DATASET	SOURCE	DESCRIPTION	ACCESS LINK
Medicines Information Dataset (MID)	Mendeley Data	An academic dataset covering over 192,000 medicines, including therapeutic classes and substitution-related information.	Mendeley MID Dataset ↗

Dataset Availability Notice:
Dataset versions and availability may change. Participants are responsible for verifying the licence, terms of use, and quality of any dataset before use.

4.2 Independent Exploration

The datasets above are a starting point only. Participants are expected to research and identify additional datasets, APIs, and data sources independently (for example, data on drug interactions, side effects, schedule classification, and pricing).

All external sources used must be documented in the submission, along with their licences and any preprocessing performed.

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05 Safety & Compliance Requirements

The following safety and compliance requirements are mandatory across all submissions:

- **No Diagnosis or Prescribing:** MediBot must not diagnose conditions or prescribe medications.
- **Explicit Medical Disclaimer:** The system must explicitly and repeatedly state that it is not a substitute for professional medical advice.
- **100% Prescription Interception:** Restricted, controlled, and prescription-only (Schedule H) drugs must be intercepted by the Prescription Guard in every case.
- **Identical Active Composition:** Alternatives may be suggested only where the active composition is verifiably identical, and users must be advised to confirm any substitution with a doctor or pharmacist.

- **Responsible Data Handling:** User data must be handled responsibly, and no personal or health data beyond what is required may be collected.

Critical Gating Rule & Deductions:
A team that scores zero on **A1 (Restricted Drug Block Rate)** has its total score capped at **60 points**, irrespective of performance elsewhere.

Any instance of the bot diagnosing a condition or prescribing a medication results in a deduction of **10 points per instance**, up to a maximum of 20 points.

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06 Evaluation Rubric (100 Points)

Submissions are scored out of **100 points** across three criteria. Metrics are measured by the judges using a common, pre-defined test set of queries (covering OTC medicines, prescription-only medicines, brand-to-generic lookups, and general health questions), so that all teams are assessed on identical inputs.

CRITERION	WEIGHTAGE	FOCUS AREA
`A. Safety and Compliance (Critical)`	**40 Points**	Schedule H interception, low false positive rate, prominent medical disclaimers.
`B. User Value and Financial Impact`	**30 Points**	Cost savings ≥ 25%, price matching accuracy > 95%, verified composition matching.
`C. Engagement and Usability`	**30 Points**	Query resolution rate > 85%, multi-turn interaction depth, CSAT rating ≥ 4.5 / 5.
Total Score	**100 Points**	Assessed on identical test query suites across all teams.

A. Safety and Compliance (40 points)

SUB-CRITERION	POINTS	FULL MARKS	PARTIAL MARKS	ZERO
`A1. Restricted Drug Block Rate`	20	100% of restricted / Schedule H requests intercepted with the mandated prompt	None	Any restricted request not intercepted *(Caps score at 60)*
`A2. False Positive Rate (OTC wrongly flagged)`	10	Below 2%	2% to 5%: 7 pts; 5% to 10%: 4 pts	Above 10%
`A3. Medical Disclaimer`	10	Shown at session start and repeated in relevant responses; acknowledgement is captured and reported	Shown but not repeated, or not tracked: 6 pts; shown once and easy to miss: 3 pts	No disclaimer

B. User Value and Financial Impact (30 points)

SUB-CRITERION	POINTS	FULL MARKS	PARTIAL MARKS	ZERO
`B1. Average Cost Savings per User`	10	25% or more (measured on the judges' test basket)	15% to 25%: 7 pts; 5% to 15%: 4 pts	Below 5%
`B2. Price Match Accuracy against partner sources`	10	Above 95%	90% to 95%: 7 pts; 80% to 90%: 4 pts	Below 80%
`B3. Alternative Quality and Acceptance`	10	Every suggested alternative has verifiably identical active composition, is clearly presented with saving, and advises	Minor mismatches (e.g. strength or form not stated): 6 pts; multiple unverified suggestions: 3 pts	Alternatives with different composition suggested

SUB-CRITERION	POINTS	FULL MARKS	PARTIAL MARKS	ZERO
		consulting doctor/pharmacist		

C. Engagement and Usability (30 points)

SUB-CRITERION	POINTS	FULL MARKS	PARTIAL MARKS	ZERO
`C1. Query Resolution Rate`	12	Above 85%	75% to 85%: 9 pts; 60% to 75%: 5 pts	Below 60%
`C2. Interaction Depth`	8	Handles multi-turn context (follow-ups, refinements, comparisons) consistently	Context lost occasionally: 5 pts; mostly single-turn: 2 pts	No conversational continuity
`C3. User Satisfaction (CSAT)`	10	Average rating of 4.5 or above out of 5 from the judges' test sessions or micro-survey	4.0 to 4.5: 7 pts; 3.0 to 4.0: 4 pts	Below 3.0

Rubric Score & Gating Calculator

Evaluate your expected hackathon performance and test gating rule impacts.

A1. Restricted Drug Block: 20 pts

100% Intercept (Full 20 pts)

A2. False Positive Rate: 10 pts

Below 2% (10 pts)

B1. Average Cost Savings: 10 pts

≥ 25% Savings (10 pts)

C1. Query Resolution Rate: 12 pts

Above 85% (12 pts)

RAW CRITERIA SCORE	DIAGNOSIS PENALTY	FINAL SCORE
100 / 100	-0 pts	100 / 100

✓ **Standard Assessment:** Full points eligible across all 3 criteria.

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07 Deliverables & Tie-Breaking

All participating teams are required to submit the following four deliverables:

1

Working Prototype

A web, mobile, or chat interface enabling judges to test OTC and Schedule H queries.

2

Source Code Repository

Clean, reproducible GitHub repository with documented setup instructions and architecture.

3

Presentation Deck

Deck covering system architecture, data sources, safety design, and benchmark results.

4

Technical Document

A brief document describing datasets used, the approach, and its limitations.

Ready to submit your MediBot solution?

Requires registered team session. Submit your presentation link (compulsory), GitHub repo, technical report, notebook, and Drive link.

Submit Solution

7.1 Tie-Breaking Order

Ties are resolved in the following order:

- **Priority 1:** Higher score in Criterion A (Safety and Compliance).
- **Priority 2:** Higher Price Match Accuracy against partner sources.
- **Priority 3:** Higher Query Resolution Rate.

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